The ear's insensitivity to phase can be understood by examining how sound propagates through the environment. Suppose you are listening to a person speaking across a small room. Much of the sound reaching your ears is reflected from the walls, ceiling and floor. Since sound propagation depends on frequency (such as: attenuation, reflection, and resonance), different frequencies will reach your ear through different paths. This means that the relative phase of each frequency will change as you move about the room. Since the ear disregards these phase variations, you perceive the voice as *unchanging* as you move position. From a physics standpoint, the phase of an audio signal becomes randomized as it propagates through a complex environment. Put another way, the ear is insensitive to phase because it contains little useful information.

However, it cannot be said that the ear is completely deaf to the phase. This is because a phase change can rearrange the *time sequence* of an audio signal. An example is the chirp system (Chapter 11) that changes an impulse into a much longer duration signal. Although they differ only in their phase, the ear can distinguish between the two sounds because of their difference in duration. For the most part, this is just a curiosity, not something that happens in the normal listening environment.

Suppose that we ask a violinist to play a note, say, the *A* below middle *C*. When the waveform is displayed on an oscilloscope, it appear much as the sawtooth shown in Fig. 22-3a. This is a result of the sticky rosin applied to the fibers of the violinist's bow. As the bow is drawn across the string, the waveform is formed as the string sticks to the bow, is pulled back, and eventually breaks free. This cycle repeats itself over and over resulting in the sawtooth waveform.

Figure 22-3b shows how this sound is perceived by the ear, a frequency of 220 hertz, plus harmonics at 440, 660, 880 hertz, etc. If this note were played on another instrument, the waveform would *look* different; however, the ear would still hear a frequency of 220 hertz plus the harmonics. Since the two instruments produce the same fundamental frequency for this note, they sound similar, and are said to have identical *pitch*. Since the relative amplitude of the *harmonics* is different, they will not sound identical, and will be said to have different *timbre*.

It is often said that timbre is determined by the shape of the waveform. This is true, but slightly misleading. The perception of timbre results from the ear detecting harmonics. While harmonic content is determined by the shape of the waveform, the insensitivity of the ear to phase makes the relationship very one-sided. That is, a particular waveform will have only one timbre, while a particular timbre has an infinite number of possible waveforms.

The ear is very accustomed to hearing a fundamental plus harmonics. If a listener is presented with the combination of a 1 kHz and 3 kHz sine wave, they will report that it sounds natural and pleasant. If sine waves of 1 kHz and 3.1 kHz are used, it will sound objectionable.